

Keshav Singh

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PROFILE

DevOps Engineer with 4+ years of professional experience in deployment automation, cloud infrastructure, and containerized applications. Hands-on skills in AWS, Terraform, Jenkins, Docker, and Kubernetes, supported by independent projects in cost optimization, troubleshooting, and observability. AWS Certified DevOps Engineer – Professional with an MSc in Cloud Computing.

EXPERIENCE

DevOps Engineer

Neosoft Systems & Cloud Services

Jan 2025 – Aug 2026

Pune, India

- Managed and enhanced CI/CD workflows using Jenkins and GitHub Actions, automating build, testing, and deployment processes and reducing manual release effort by approximately 40%.
- Provisioned and maintained AWS infrastructure using Terraform across EC2, VPC, S3, IAM, RDS, and load-balancing services, reducing environment provisioning time by approximately 50%.
- Supported production and non-production workloads on Docker and Kubernetes, managing deployments, configuration, resource limits, health probes, and rolling updates across multiple application environments.
- Developed Python and Bash automation scripts for deployment validation, log collection, environment checks, and routine operational tasks, saving approximately 8–10 hours of manual effort per month.
- Improved application and infrastructure observability using CloudWatch, Prometheus, and Grafana, creating dashboards and alerts that helped reduce troubleshooting and incident-response time by approximately 25%.
- Supported releases across development, testing, and production environments, performing pre-deployment checks, post-release validation, troubleshooting failed deployments, and coordinating rollback procedures.
- Strengthened cloud and Kubernetes access controls using AWS IAM, Kubernetes RBAC, secrets management, and least-privilege permissions to support secure application deployments.
- Investigated Linux, networking, AWS, and container-related incidents using system logs, Kubernetes diagnostic tools, and CloudWatch metrics, while maintaining operational runbooks for recurring issues.

Software Engineer

Zensar Technologies

Nov 2020 – Mar 2023

Pune, India

- Designed and maintained 25+ CI/CD pipelines, reducing deployment time from 45 minutes to under 10 minutes.
- Automated AWS and Azure environment provisioning using Terraform, reducing infrastructure setup time by 60%.
- Supported containerized microservices on Kubernetes using Helm, autoscaling, health probes, and controlled deployment strategies.
- Implemented access and deployment security controls, including IAM, RBAC, secrets management, and vulnerability scanning.
- Troubleshoot build, deployment, and infrastructure incidents, achieving a 98% deployment success rate and reducing mean time to recovery by 30%.
- Built monitoring and alerting using Prometheus, Grafana, CloudWatch, and Azure Monitor for 20+ services, improving incident detection and operational visibility.

PROJECTS

CloudSpend-Guard – AWS Cost Optimization

Aug 2026 – Present

Independent Project – GitHub Repository

AWS / Python / DevOps

- Built a platform that discovers EC2 and EBS resources, analyzes CloudWatch utilization and AWS pricing data, and generates cost-saving recommendations.
- Developed a React and TypeScript dashboard with Vite and a FastAPI backend using boto3; tested resource discovery and recommendation APIs with pytest.
- Provisioned infrastructure using Terraform and deployed containerized workloads to private Amazon EKS using Docker, Helm, ECR, VPC networking, and an ALB.
- Built GitHub Actions workflows for testing, image builds, and ECR publishing, using GitHub OIDC, immutable Git-SHA image tags, and least-privilege IAM.

KubeDiagnose – Kubernetes Diagnostics CLI

Independent Project – GitHub Repository

Kubernetes / Troubleshooting

- Built a Python command-line tool to diagnose Kubernetes workload issues, with workload and namespace selection through the `diagnose` command.
- Created demonstration workloads for CrashLoopBackOff, image pull failures, out-of-memory errors, failed probes, and scheduling issues to exercise troubleshooting scenarios.
- Documented command usage and diagnostic examples with screenshots to demonstrate investigation of common Kubernetes failures.

Telemetry-Lab – Observability Project

Independent Project – GitHub Repository

Observability

In Development – Planned Stack; Implementation Details to Be Confirmed.

- Defined an observability lab for application metrics, logs, and traces, with a planned stack of OpenTelemetry, Prometheus, Grafana, Loki, and Tempo.
- Planned a FastAPI application with PostgreSQL, Docker Compose, and Kubernetes to demonstrate the observability workflow.

CORE SKILLS

Cloud & Networking: AWS (EC2, S3, IAM, VPC, EKS, ECR, ALB), Azure, GCP

Infrastructure as Code: Terraform, CloudFormation, AWS CDK

CI/CD & Release Automation: Jenkins, GitHub Actions, GitLab CI/CD, Azure DevOps, AWS CodePipeline

Containers & Orchestration: Docker, Kubernetes, Amazon EKS, Helm

Monitoring & Logging: Prometheus, Grafana, CloudWatch, Azure Monitor

Programming Languages: Python, Java, TypeScript, SQL

Frontend: React, Vite

Backend & APIs: FastAPI, boto3, REST APIs

Testing & Quality: pytest, Automated Testing, Linting

Scripting & Automation: Bash, Ansible, AWS Systems Manager

Operating Systems & Version Control: Linux, Git

Security & Access Management: IAM, RBAC, AWS Secrets Manager, GitHub OIDC, EKS Pod Identity

CERTIFICATION

AWS Certified DevOps Engineer – Professional

EDUCATION

National College of Ireland

MSc in Cloud Computing

Dublin, Ireland

2023 – 2024